Universidad San Carlos de Guatemala

Lab: Redes 1

Sección A

Tarea 4

Configuración de Routers

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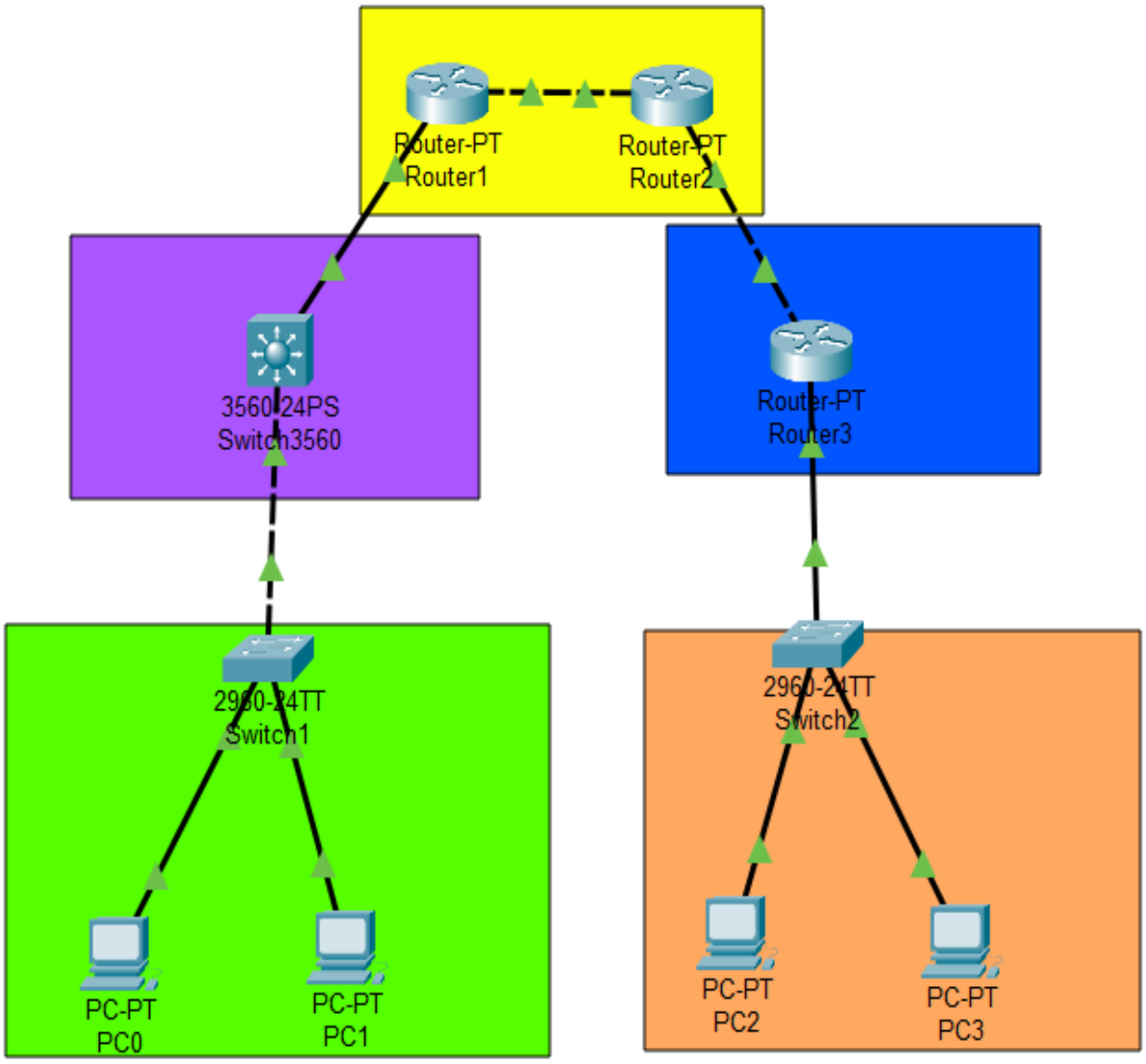
TOPOLOGIA


Router-PT
Router1
Router-PT
Router2
3560-24PS
Switch3560
Router-PT
Router3
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3

ENRUTAMIENTOS DE CADA ROUTER

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!


User Access Verification

Password:

Router1>en
Password:
Router1#how ip route
            ^
% Invalid input detected at '^' marker.

Router1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
C       10.0.1.0 is directly connected, FastEthernet0/0
C       10.0.2.0 is directly connected, FastEthernet1/0
O       10.0.3.0 [110/2] via 10.0.2.2, 01:48:35, FastEthernet1/0
     192.168.33.0/25 is subnetted, 2 subnets
R       192.168.33.0 [120/1] via 10.0.1.2, 00:00:14, FastEthernet0/0
R       192.168.33.128 [120/1] via 10.0.1.2, 00:00:14, FastEthernet0/0
     192.168.34.0/25 is subnetted, 2 subnets
O E2    192.168.34.0 [110/20] via 10.0.2.2, 01:37:40, FastEthernet1/0
O E2    192.168.34.128 [110/20] via 10.0.2.2, 01:37:40, FastEthernet1/0

Router1#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
User Access Verification

Password:
Password:

Router2>en
Password:
Password:
Router2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
O E2    10.0.1.0 [110/20] via 10.0.2.1, 01:41:07, FastEthernet1/0
C       10.0.2.0 is directly connected, FastEthernet1/0
C       10.0.3.0 is directly connected, FastEthernet0/0
     192.168.33.0/25 is subnetted, 2 subnets
O E2    192.168.33.0 [110/20] via 10.0.2.1, 01:41:07, FastEthernet1/0
O E2    192.168.33.128 [110/20] via 10.0.2.1, 01:41:07, FastEthernet1/0
     192.168.34.0/25 is subnetted, 2 subnets
D       192.168.34.0 [90/30720] via 10.0.3.2, 00:05:40, FastEthernet0/0
D       192.168.34.128 [90/30720] via 10.0.3.2, 00:05:40, FastEthernet0/0

Router2#
```

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Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.3.1 (FastEthernet1/0) is down: holding time expired

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.3.1 (FastEthernet1/0) is up: new adjacency

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.3.1 (FastEthernet1/0) is down: Interface Goodbye
received

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.3.1 (FastEthernet1/0) is up: new adjacency


User Access Verification

Password:

Router3>en
Password:
Password:
Router3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/30 is subnetted, 3 subnets
D EX    10.0.1.0 [170/284160] via 10.0.3.1, 00:07:04, FastEthernet1/0
D       10.0.2.0 [90/30720] via 10.0.3.1, 00:07:04, FastEthernet1/0
C       10.0.3.0 is directly connected, FastEthernet1/0
     192.168.33.0/25 is subnetted, 2 subnets
D EX    192.168.33.0 [170/284160] via 10.0.3.1, 00:07:04, FastEthernet1/0
D EX    192.168.33.128 [170/284160] via 10.0.3.1, 00:07:04, FastEthernet1/0
     192.168.34.0/25 is subnetted, 2 subnets
C       192.168.34.0 is directly connected, FastEthernet0/0.30
C       192.168.34.128 is directly connected, FastEthernet0/0.40

Router3#
```

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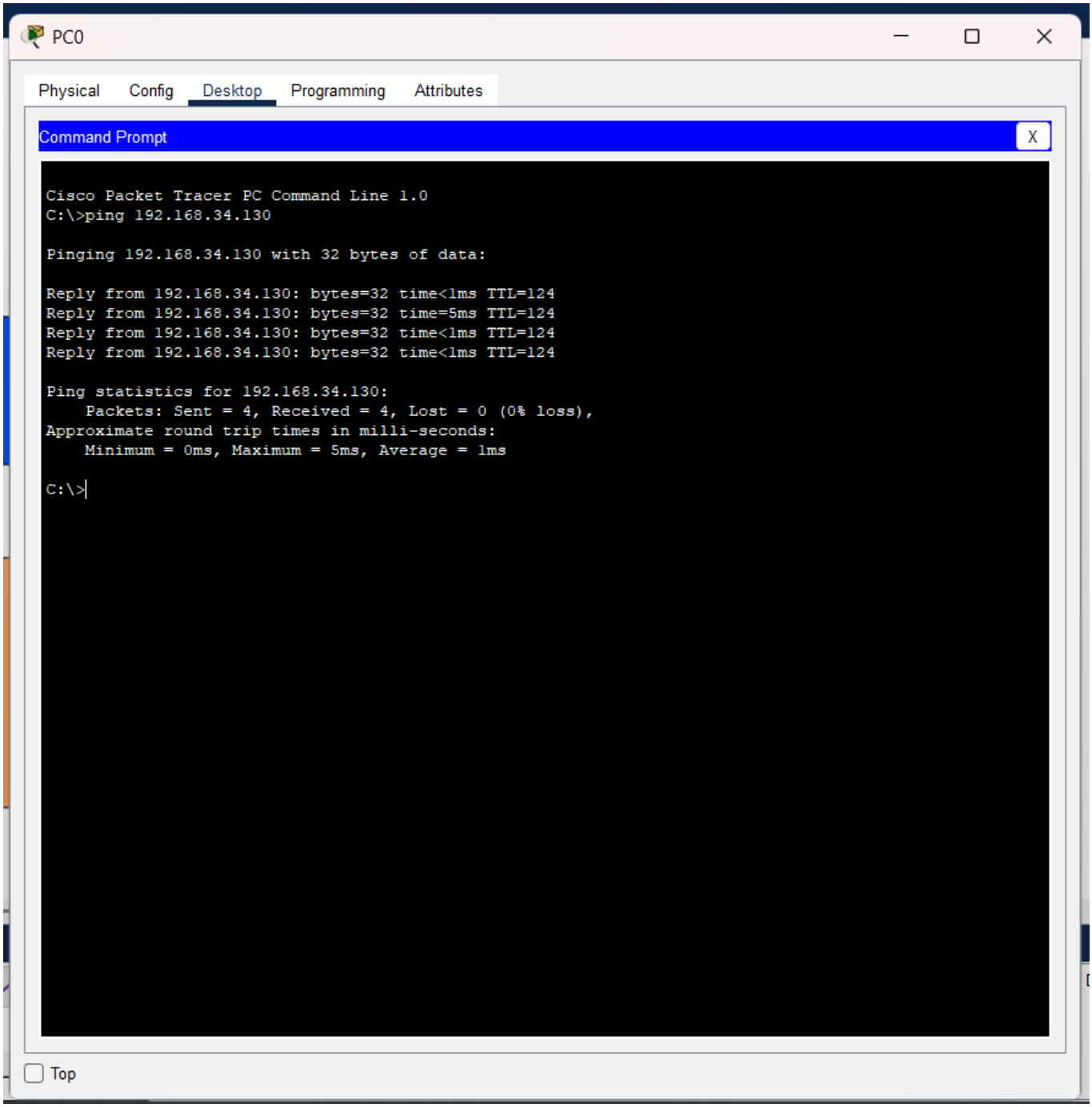
Ping entre PC0 y PC3



PREGUNTAS

a. Reflexión

Con esta tarea aprendí que configurar varios protocolos de enrutamiento juntos no es tan difícil si lo haces paso a paso. Al principio me confundí con la redistribución porque no entendía por qué los pings fallaban aunque todo parecía bien. Fui probando comando por comando hasta encontrar dónde estaba el error. Aprendí que equivocarse es parte del proceso.

b. Aplicación práctica

En mi vida profesional esto me va a servir mucho porque en las empresas no todos los equipos hablan el mismo protocolo de enrutamiento. Si algún día trabajo conectando redes de diferentes sucursales o migrando infraestructura, voy a saber cómo hacer que se comuniquen aunque usen protocolos distintos. Básicamente aprendí a hacer que redes "incompatibles" se entiendan entre sí.

c. Conclusión

Esta tarea me hizo configurar RIP, OSPF y EIGRP al mismo tiempo, algo que nunca había hecho antes. Lo más interesante fue ver cómo la redistribución es la pieza clave que une todo, sin ella ningún protocolo sabe lo que el otro conoce. Al final logré conectividad completa entre PC0 y PC3, lo que me demostró que la teoría vista en clase sí funciona en la práctica.